AI-Based Detection and Analysis of Dark Patterns

A Human-Validated India-Focused Text Baseline with a CCPA-Aware Evaluation Framework

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Abstract. Dark patterns are deceptive interface practices that steer users toward outcomes that benefit the platform. We evaluate an India-focused text detector against an independently held-out gold set of 499 human-labelled Indian interface snippets, of which 26 are dark patterns. The gold set is never used for training. A soft-voting text model achieves an estimated pool-weighted F1 of 0.171 (95% bootstrap range: 0.034-0.493), compared with 0.056 (0.019-0.109) for a pseudo-labeller baseline, and outperforms the baseline in 95% of paired resamples. At a 0.50 threshold, the model produces 11 false alarms, versus 155 for the baseline, but detects only 9 of 26 dark rows. The large gap from an in-distribution F1 of 0.927 to the human-labelled Indian evaluation demonstrates the importance of domain-specific validation. We provide category results, a CCPA category crosswalk, error analysis, and ethical limitations. Multimodal fusion and transformer comparisons remain future work; no performance claims are made for either.

Keywords: dark patterns; deceptive design; human evaluation; CCPA; Indian e-commerce; domain shift; machine learning

1. Introduction

Dark patterns use interface wording, layout, or interaction flow to influence users in ways that primarily benefit the platform. Examples include false urgency, basket sneaking, confirm shaming, and subscription traps. India's Central Consumer Protection Authority (CCPA) published the Guidelines for Prevention and Regulation of Dark Patterns in 2023, creating a practical need for evidence-based identification of such practices [5].

Automated dark-pattern research has used text, visual signals, DOM structure, and mobile-app exploration [1-4]. However, strong scores on benchmark corpora may not transfer to Indian user interfaces or to naturally imbalanced, human-labelled data. This revision therefore narrows the completed contribution to a transparent text evaluation and makes the main empirical result the measured domain gap. The previously proposed multimodal framework is retained only as a future research design.

Contributions

(1) We evaluate a text detector on a held-out, human-labelled Indian gold set of 499 snippets; (2) we compare it with a pseudo-labeller baseline using pool-weighted estimates and paired bootstrap resampling; (3) we report category performance, error patterns, and a CCPA crosswalk; and (4) we document constraints that prevent legal or multimodal claims.

2. Related Work

Bajaj et al. [1] combine structural, visual, and textual signals; AidUI [2] localizes UI patterns with computer vision and NLP; UIGuard [3] uses knowledge-driven checking for mobile UIs; and AppRay [4] explores dynamic app states. These systems motivate multimodal research, but their task definitions, platforms, and datasets differ from the present text-only evaluation. Mathur et al. [6] established web-scale evidence of dark patterns, while Yada et al. [7] released a text dataset and baseline evaluations. Our result is not a claim of a superior universal detector; it measures how a model trained with available data behaves on held-out Indian text.

3. Data, Labels, and CCPA Crosswalk

The in-distribution training corpus includes public ec-darkpattern text records derived from the dataset of Yada et al. [7], itself based on dark-pattern texts reported by Mathur et al. [6] with additional non-dark-pattern text from the same e-commerce sites. The original benchmark contains 2,356 samples: 1,178 dark-pattern and 1,178 non-dark-pattern texts. Each record contains an identifier, interface text, binary label, and fine-grained category. The 50/50 class balance is suitable for controlled benchmarking but is not representative of real deployment prevalence.

The external gold set contains 499 Indian snippets manually labelled during collection and is held out from training and model selection. Twenty-six rows are marked as dark patterns. To estimate performance across the roughly 26,000 scraped snippets represented by the sampling design, each gold row is weighted by the number of snippets it represents. Because dark examples are rare, the 95% bootstrap ranges are wide and small differences should not be overinterpreted.

Annotation limitation. The current gold labels were produced by a single annotator. Consequently, this revision cannot report inter-annotator agreement. Before a future release or legal-facing deployment, an independently double-labelled subset, a written codebook, and agreement statistics (Cohen's kappa or Krippendorff's alpha) are required.

Pattern family	CCPA interpretation	Retained evidence
Urgency / Scarcity	False Urgency	Text span, timer or stock phrase
Social Proof	False or misleading popularity claim	Popularity claim and surrounding context
Misdirection	Confirm Shaming, Interface Interference, or Trick Question	Exact wording and UI context
Obstruction	Subscription Trap	Cancellation-flow text
Sneaking	Basket Sneaking or Drip Pricing	Checkout text and price disclosure
Forced Action	Forced Action	Required action or disclosure

Table 1. CCPA-oriented interpretation crosswalk. This is an analytic mapping, not a legal finding.

4. Methods

The completed detector uses word and character n-gram TF-IDF features with a soft-voting ensemble of LightGBM, support-vector, and logistic-regression classifiers for binary detection. Model selection uses cross-validation on the training data only. A separate logistic-regression classifier predicts a reduced category label set; rare Sneaking and Forced Action samples are merged into Other because their available support is too small for stable standalone estimates.

Figure 1 separates completed work from the planned multimodal extension. The screenshot branch, fusion mechanism, and CCPA category validation have not been implemented or evaluated in this study. Likewise, transformer models were not trained for this revision and are not used as a comparison basis.

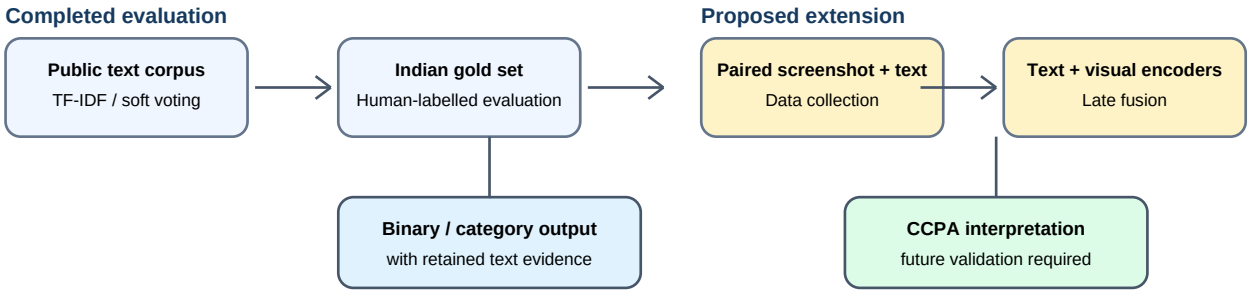

Figure 1. Completed text evaluation (blue) and proposed multimodal extension (amber). The proposed components are not reported as experimental results.

5. Experimental Protocol

In-distribution results use a stratified held-out 20% split of the training corpus. External evaluation uses the 499-row Indian gold set only once after model selection. We report pool-weighted precision, recall, and F1, plus row-level counts to make the impact of the sampling design visible. Uncertainty is estimated through paired bootstrap resampling of the gold evaluation. The baseline is a pseudo-labeller, not a supervised model trained against the gold labels.

System	Precision	Recall	F1 [95% range]	Caught	False alarms
Soft voting, threshold 0.50	0.410	0.108	0.171 [0.034-0.493]	9/26	11
Soft voting, CV-tuned 0.18	0.113	0.159	0.132 [0.046-0.260]	16/26	75
Pseudo-labeller baseline	0.032	0.238	0.056 [0.019-0.109]	12/26	155

Table 2. Pool-weighted external evaluation on the held-out Indian gold set (n = 499; 26 dark rows).

6. Results and Discussion

The main result is an improvement over the pseudo-labeller baseline at the default threshold: estimated F1 rises from 0.056 to 0.171, a +0.115 difference (95% range: -0.012 to +0.432), and the model has higher F1 in 95% of paired resamples. The gain is driven primarily by precision: false alarms fall from 155 to 11. The trade-off is low recall. At the 0.50 threshold, the model identifies 9 of 26 dark rows; lowering the threshold identifies 16 but creates 75 false alarms. These results are too uncertain to support claims of small improvements, but they clearly show that in-distribution performance does not establish Indian deployment performance.

On the in-distribution held-out split, soft voting obtains cross-validation F1 = 0.913, test F1 = 0.927, and accuracy = 0.966. The category model obtains test F1 = 0.956 and accuracy = 0.955 (macro-F1 = 0.90); such results reflect the benchmark distribution, not the gold set. On the 26 dark gold rows, the category model achieves 0.62 accuracy and macro-F1 = 0.57, a far more tentative estimate due to small support.

Evaluation	F1 / macro-F1	Accuracy	Interpretation
Binary, benchmark split	0.927	0.966	In-distribution only
Category, benchmark split	0.90 macro	0.955	Rare labels merged
Binary, Indian gold	0.171 weighted	Not prevalence-informative	External, low-prevalence evaluation
Category, 26 gold dark rows	0.57 macro	0.62	Exploratory; very small support

Table 3. In-distribution and external results must not be compared as if they were the same task.

Error analysis

Most remaining false negatives involve subtle or contextual claims that are lexically unlike benchmark examples, such as “Trusted by 20+ million customers.” The model also misses novel numeric urgency strings. False positives are substantially reduced but remain possible when neutral inventory or promotional language resembles scarcity. The observed external domain gap indicates that lexical overlap is insufficient to establish deceptive intent; surrounding UI context and independent human review remain necessary.

7. Ethics, Legal Scope, and Data Availability

Collection should be limited to public interface content, rate limited, and designed to minimize capture of personal data. Screenshots and text should be redacted where they contain identifiers. Predictions may cause harm if they wrongly label a legitimate business practice as deceptive; outputs must therefore retain evidence and be reviewed by a human. The CCPA mapping is a research-oriented interpretation of guideline categories, not legal advice, an enforcement finding, or a determination of liability.

We intend to release permissible de-identified annotations, label definitions, model code, split identifiers, and the CCPA crosswalk. Raw screenshots or text that cannot be redistributed because of platform terms, copyright, privacy, or security concerns will not be released; reproducible collection instructions and derived annotations should be provided instead.

8. Limitations and Future Work

This work does not yet include a transformer baseline, a fully implemented screenshot branch, multimodal fusion, or site-disjoint cross-site testing. The gold data has a small number of positive cases and was singly annotated. The reported confidence ranges are consequently broad. Future work should create a double-annotated, site-disjoint Indian dataset; compare transformers and visual models under the same protocol; validate CCPA assignments; and evaluate calibrated human-in-the-loop operating points.

9. Conclusion

A strong benchmark score does not transfer automatically to real Indian interface text. Although the model improves pool-weighted F1 over the pseudo-labeller baseline and sharply reduces false alarms, it still misses most dark examples. The appropriate conclusion is not that dark-pattern detection is solved, but that human-labelled, domain-specific evaluation exposes limitations masked by balanced benchmark splits. The proposed multimodal and CCPA-aware system remains a testable next step, not a completed result.

References

[1] A. Bajaj et al., “A Complete Analysis for Dark Pattern Detection Using Structural, Visual and Textual Information,” IJCISIM, 17, 14-25, 2025. [2] S. M. H. Mansur et al., “AidUI,” ICSE, 1958-1970, 2023. [3] J. Chen et al., “Unveiling the Tricks: Automatic Detection of Dark Patterns in Mobile Applications,” UIST, 2023. [4] J. Chen et al., “From Exploration to Revelation: Detecting Dark Patterns in Mobile Apps,” arXiv:2411.18084, 2024. [5] Central Consumer Protection Authority, Guidelines for Prevention and Regulation of Dark Patterns, 2023. [6] A. Mathur et al., “Dark Patterns at Scale,” PACM HCI, 3(CSCW), 2019. [7] Y. Yada et al., “Dark Patterns in E-Commerce: A Dataset and Its Baseline Evaluations,” IEEE Big Data, 2022.

Appendix A. Point-by-Point Response to Reviewers

This report is appended to the revised manuscript as requested. Page references correspond to the revised PDF. Items requiring additional data, notably transformer/multimodal testing and inter-annotator agreement, are transparently identified as remaining limitations rather than presented as completed evidence.

Reviewer 1 - Low novelty of TF-IDF logistic regression

Addressed on pp. 1-3. The paper now describes the classical text method as a baseline, not a novel algorithm. The main contribution is the external human-labelled Indian evaluation and the observed domain gap.

Reviewer 1 - Source and labels of 2,356 samples

Addressed on p. 1. The public source, four fields, class balance, provenance, and role of the benchmark corpus are specified.

Reviewer 1 - Transformer and existing-detector comparison

Partly addressed on pp. 1 and 3. Prior detector systems are compared as related work, with modality/task differences stated. A matched transformer experiment is not yet available and is explicitly listed as future work; no competitiveness claim is made.

Reviewer 1 - Implement multimodal framework

Addressed by scope correction on pp. 1-2 and 3. Figure 1 is labelled proposed. The manuscript removes performance claims for screenshot fusion or multimodal detection.

Reviewer 2 - Expand error analysis

Addressed on p. 2. The revision explains false-alarm reduction, low recall, and representative wording patterns behind remaining errors.

Reviewer 2 - Show CCPA connection

Addressed on p. 2 in Table 1. The crosswalk links pattern families, CCPA interpretations, and retained evidence, with a non-legal-advice qualifier.

Reviewer 2 - Cross-site generalisation

Not yet fully implemented; stated on p. 3. The present revision provides a stronger external Indian evaluation but not a leave-one-site-out protocol. This is retained as a specific next experiment.

Reviewer 2 - Balanced data unrealistic

Addressed on pp. 1-2. The manuscript explains why the balanced corpus cannot represent prevalence and reports low-prevalence gold-set evaluation.

Reviewer 2 - Per-category results

Addressed on p. 2 in Table 3. Benchmark and gold-category results are reported with explicit cautions about rare-category support.

Reviewer 2 - Ethical/legal issues

Addressed on p. 3. A dedicated section covers public collection, privacy, false-positive harm, human review, data sharing, and CCPA legal scope.

Reviewer 3 - Framework diagram

Addressed on p. 2 in Figure 1, which distinguishes completed and proposed components.

Reviewer 3 - Annotation agreement

Not yet available; transparently disclosed on pp. 1 and 3. The paper does not fabricate agreement and defines the double-annotation protocol required for a subsequent release.

Reviewer 3 - Dataset release

Addressed on p. 3. A data-availability plan defines what can be released and how restricted material will be handled.

Reviewer 3 - Recent literature and conclusion

Addressed on pp. 1 and 3. Related work differentiates comparable and non-comparable systems; the conclusion now reflects only validated findings.